DIPLOMA PROGRAMME PRACTICE / HL / SET A

Economics / Paper 3

Question booklet · 105 minutes · 60 marks

Answer all parts of both questions. A calculator is permitted. Write your answers and diagrams in the spaces provided. Show working and units. Give exact values or round to two decimal places unless otherwise instructed.

QUESTION 1 / 30 MARKS

A levy on single-use containers

Harbour City sends discarded takeaway containers to a landfill beside a fishing inlet. A local study values the extra external damage at 6 crowns per container throughout the relevant range. Firms and consumers pay none of this damage. Each week the competitive market follows Qd = 120 − 2Pc and Qs = 20 + 2Pp, where quantities are in thousands of containers and prices are crowns per container. Pc is the buyer's price; Pp is the seller's net receipt.

The council proposes a specific tax of 10 crowns. Assume full collection, linear curves and no change in other markets for parts (a)(i)–(vii). Fishing households benefit from less waste. A reuse deposit is another option, but collecting returned containers requires new facilities in outer districts. A small weekly survey found disposable packaging was most common among outlets without washing space.

a(i) Calculate the equilibrium price before intervention. **[2]**

a(ii) Calculate equilibrium quantity and total external damage before intervention. **[2]**

a(iii) Calculate the buyer's and seller's prices after the proposed tax. **[3]**

a(iv) Calculate the new weekly quantity and tax revenue. State the share of the per-unit levy borne by buyers. **[3]**

a(v) Calculate the weekly external damage avoided under the proposed levy. **[2]**

a(vi) Using a diagram, explain why the proposed levy could reduce quantity below the socially efficient level. **[4]**

a(vii) Calculate the change in total weekly welfare from the proposed levy relative to no intervention. Treat tax revenue as a transfer. **[4]**

b Using the information and your knowledge of economics, recommend one policy for Harbour City to reduce the social cost of disposable containers. **[10]**

QUESTION 2 / 30 MARKS

An output gap and a repair programme

Bracken's statistics office estimates current real GDP at 940 billion crowns and potential real GDP at 1,000 billion, at the same prices. An income survey estimates marginal propensities to save, pay net taxes and import at 0.20, 0.10 and 0.20 respectively, all measured per additional crown of national income. Use the simple open-economy multiplier with constant marginal leakages and no crowding out for the calculations.

A bridge repair plan would add 40 billion crowns to government purchases in the first year. A separate option is a temporary transfer to low-income households, some of which would be saved. Contractors in the construction sector report material shortages even though national unemployment is high. Public debt is 600 billion crowns, and the existing annual budget deficit is 18 billion. Monetary policy and prices may respond outside the simple model.

a(i) Calculate the recessionary output gap as a percentage of potential GDP. **[2]**

a(ii) Calculate the simple expenditure multiplier. **[2]**

a(iii) Using the multiplier, calculate the predicted increase in equilibrium income and the resulting GDP if the full repair plan is an additional injection. **[3]**

a(iv) Calculate the additional government purchases that would close the initial gap in this model and compare your result with the proposed plan. **[3]**

a(v) If GDP rises by 60 billion crowns, calculate the induced increase in net tax receipts and in imports under the stated propensities. **[2]**

a(vi) Assuming a 30 billion purchase, the 6 billion increase in net tax receipts and no other budget changes, calculate the new deficit and year-end debt. Explain the distinction. **[4]**

a(vii) Using an AD–AS diagram, explain why the repair plan might raise real GDP by less than its multiplier prediction. **[4]**

b Using the information and your knowledge of economics, recommend a fiscal response to the recessionary gap in Bracken. **[10]**